Southwire®
POWER GENERATION SOLUTIONS

WIRE & CABLE FOR THE GAS POWER CAMPUS

THE PATH OF POWER

From the gas yard to the 230 kV line, every megawatt a plant makes travels by cable. This is the route, and the cable it runs on.



ACCELERATING TIME TO POWER™

SOUTHWIRE · POWER GENERATION SOLUTIONS

Stephan Hardt · stephan.hardt@southwire.com

01 — THE PATH OF POWER

EVERY MEGAWATT TRAVELS BY CABLE.

Gas arrives at the yard and burns in three H-class turbines. Each generator makes power at 21 kV; a generator breaker and isolated-phase bus carry it to the step-up transformer, and 230 kV cable and conductors take it to the grid. Along the way, miles of medium- and low-voltage cable run every pump, fan and valve that keeps the turbines turning.

4.6 MI
of cable-tray routes

4.2 MI
of duct-bank and buried cable routes

417
pieces of equipment, each with its own cable

FUEL GAS

GAS TURBINE + GENERATOR
21 kV

STEP-UP TRANSFORMER
21 / 230 kV

TO THE GRID
230 kV

SK-3X1 campus, rendered from the 3D model. Route lengths measured along modelled routes, not cable footage.

FUEL GAS · GENERATION · 21 kV · AUXILIARIES · 15 kV / 600 V · STEP-UP · 21 / 230 kV · GRID · 230 kV

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02 — ANATOMY OF A 15 KV FEEDER

ARMOR-X® OPENED UP.

In the turbine hall the 13.8 kV auxiliaries run on 15 kV ARMOR-X: three shielded EPR cores and a ground under a continuously welded aluminium armor and a red PVC jacket. Armored, it lays straight into the tray with no conduit, and it is rated for the Class I, Division 2 areas around the gas turbines.

Typical construction from the customer's ARMOR-X list (500-37 3/C CPRESS CU NL-EPR 25% T/S 1X#1 CU GW, red PVC). Confirm against current Southwire specifications. Studio render of the cable model, not a photograph.

RED PVC JACKET
Sunlight resistant, printed legend every foot

CORRUGATED ALUMINIUM ARMOR
Continuously welded: impervious, crush-resistant, MC-HL

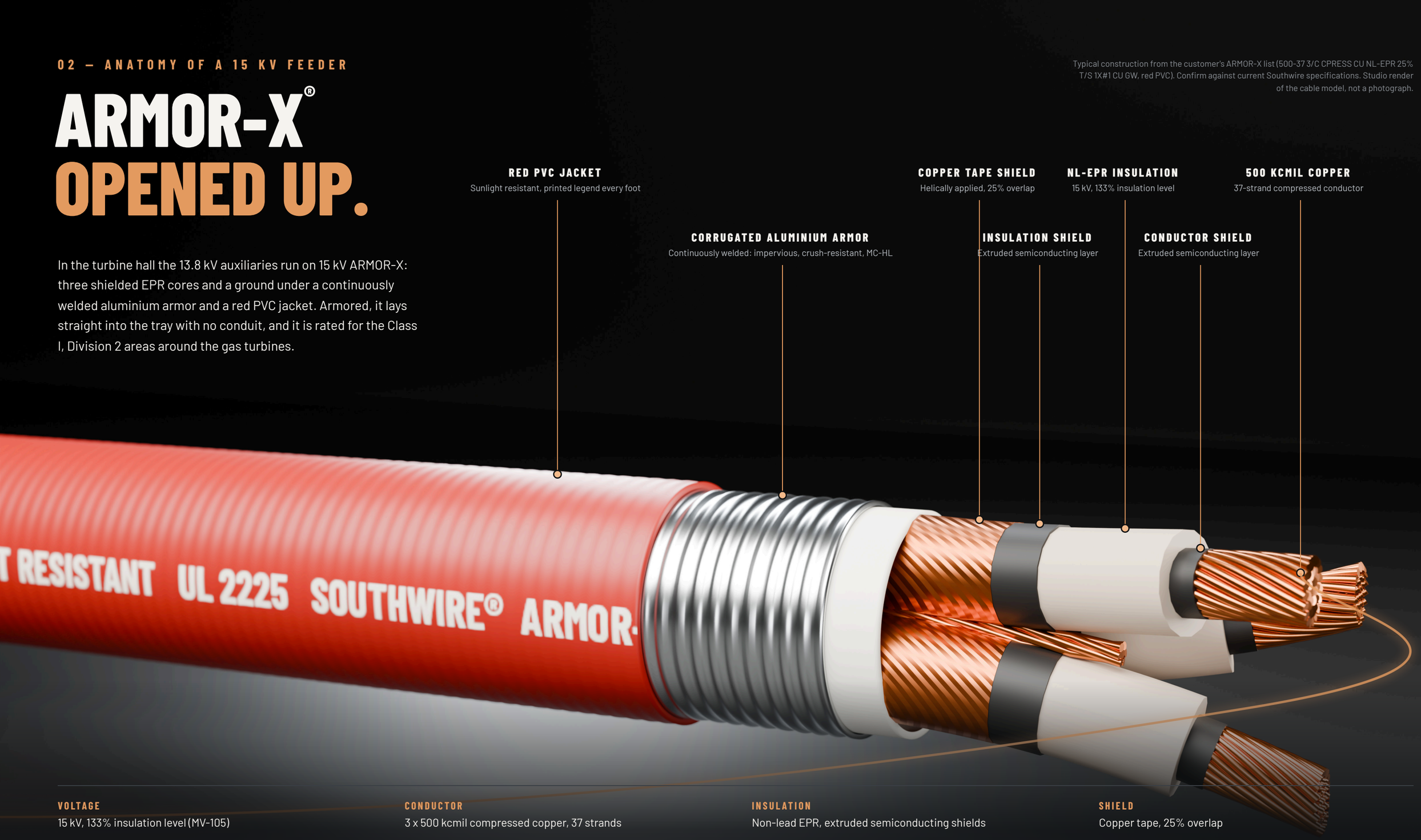
COPPER TAPE SHIELD
Helically applied, 25% overlap

INSULATION SHIELD
Extruded semiconducting layer

NL-EPR INSULATION
15 kV, 133% insulation level

CONDUCTOR SHIELD
Extruded semiconducting layer

500 KCMIL COPPER
37-strand compressed conductor



VOLTAGE
15 kV, 133% insulation level (MV-105)

CONDUCTOR
3 x 500 kcmil compressed copper, 37 strands

INSULATION
Non-lead EPR, extruded semiconducting shields

SHIELD
Copper tape, 25% overlap

GROUND
1 x #1 AWG bare copper

ARMOR
Continuously welded corrugated aluminium

JACKET
Red PVC, sunlight resistant

LISTING
Type MC-HL (UL 1569 / UL 2225) or MV-105 (UL 1072)

FUEL GAS · GENERATION · 21 kV · **AUXILIARIES · 15 kV / 600 V** · STEP-UP · 21 / 230 kV · GRID · 230 kV

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